

SE LAB 3:

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Municipal Infrastructure Damage Reporting System

1. Scenario Review

The Municipal Infrastructure Damage Reporting System allows citizens to report infrastructure damage by submitting a report with damage details and photographs. The system captures the incident location, validates GPS coordinates, automatically dispatches the report, and allows municipal personnel to review and process the damage report. Citizens can also track the status of submitted reports, while municipal personnel can update the report status and send status notifications.

2. Functional Requirements

- Citizen can submit an infrastructure damage report.
- Citizen can capture/upload damage photos and details.
- Citizen can enter the incident location.
- System validates the GPS coordinates.
- System automatically dispatches the report.
- Municipal personnel can review and process the damage report.
- Citizen can track the submitted damage report status.
- Municipal personnel can update the damage report status.
- System can send status notifications.

3. Non-Functional Requirements

- **Security:** The system should protect citizen and municipal data from unauthorized access.
- **Performance:** The system should process and dispatch damage reports efficiently.
- **Usability:** Citizens should be able to submit and track reports easily.

- **Scalability:** The system should be able to handle multiple damage reports from citizens.
- **Availability:** The system should be accessible when citizens need to report infrastructure damage.

4. Architectural Style Analysis

4.1 Layered Architecture

Structure: Presentation Layer → Business Layer → Data Layer

Advantages:

- Clear separation of concerns
- Easy to understand
- Suitable for traditional applications

Disadvantages:

- Performance overhead
- Tight coupling between layers
- Difficult to scale individual components

4.2 Microservices Architecture

Structure: Small, independently deployable services

Advantages:

- Independent scaling
- Fault isolation
- Technology diversity
- Independent deployment

Disadvantages:

- Operational complexity
- Network latency
- Data consistency challenges

4.3 Client-Server Architecture

Structure: Centralized server with multiple clients

Advantages:

- Centralized control
- Simple deployment
- Easy data consistency

Disadvantages:

- Single point of failure
- Scalability bottleneck
- Limited fault tolerance

5. Architecture Selection

Selected Architecture: Layered Architecture

Reason 1: Layered Architecture provides a clear separation of concerns between the presentation, business, and data parts of the Municipal Infrastructure Damage Reporting System.

Reason 2: It makes the system easier to understand and maintain because each layer has a specific responsibility.

6. Component Identification

- Citizen Interface Component
- Engineer/Admin Interface Component
- Damage Report Management Component
- GPS Validation Service Component
- Dispatch & Notification Service Component
- Municipal Database Component

7. Component Layer Allocation

Presentation Layer:

- Citizen Interface Component
- Engineer/Admin Interface

Component Business Layer:

- Damage Report Management Component
- GPS Validation Service Component
- Dispatch & Notification

Service Component Data Layer:

- Municipal Database Component

8. Interface Identification

- **Report Submission Interface**

Connects **Citizen Interface Component** with **Damage Report Management Component**.

Purpose: Submit the damage report, photo and details.

- **GPS Validation Interface**

Connects **Damage Report Management Component** with **GPS Validation Service Component**.

Purpose: Validate the incident GPS coordinates.

- **Dispatch Interface**

Connects **Damage Report Management Component** with **Dispatch & Notification Service Component**.

Purpose: Automatically dispatch the damage report.

- **Database Interface**

Connects **Damage Report Management Component** with **Municipal Database**

Component.

Purpose: Store and retrieve damage report information.

- **Report Management Interface**

Connects **Engineer/Admin Interface Component** with **Damage Report Management Component**.

Purpose: Allow municipal personnel to review reports and update their status.

- **Notification Interface**

Connects **Dispatch & Notification Service Component** with **Citizen Interface Component**.

Purpose: Send report-status notifications to citizens.

9. Provided and Required Interfaces

<u>Interface</u>	<u>Provided by</u>	<u>Required by</u>
Report Submission Interface	Damage Report Management Component	Citizen Interface Component
GPS Validation Interface	GPS Validation Service Component	Damage Report Management Component
Dispatch Interface	Dispatch & Notification Service Component	Damage Report Management Component
Database Interface	Municipal Database Component	Damage Report Management Component
Report Management Interface	Damage Report Management Component	Engineer/Admin Interface Component
Notification Interface	Dispatch & Notification Service Component	Citizen Interface Component

10. Written Justification

Architecture Selection:

We chose **Layered Architecture** for the Municipal Infrastructure Damage Reporting System.

Reason 1:

Layered Architecture provides clear separation of concerns between the presentation, business, and data layers. This suits the system because citizen and engineer interfaces

are separated from report processing, GPS validation, dispatch, notification, and database operations.

Reason 2:

The architecture makes the system easier to understand, test, and maintain because each layer has a specific responsibility. Changes to the user interface or database can be made with minimal impact on the other layers.

Security Advantage:

Layer separation helps control access to municipal data by ensuring that users interact through the presentation and business layers before data is accessed. This makes it easier to apply authentication, authorization, and validation at appropriate boundaries.

Performance Benefit:

Separating presentation, business processing, and data access provides clear processing paths for operations such as GPS validation, report processing, and dispatch. This helps the system process and dispatch damage reports efficiently.